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Inventor(s)

Hu; Haitian et al.

PRESSURE SENSING ASSEMBLY FOR A DISHWASHER APPLIANCE

Abstract

A dishwasher appliance includes a wash tub and a sump for collecting wash fluid, a pump assembly for selectively urging a flow of the wash fluid from the sump, a pressure sensor for monitoring a wash fluid level within the wash tub, a microphone for monitoring sound generated during operation of the dishwasher appliance, and a controller configured to obtain a pressure reading using the pressure sensor, determine that the wash fluid in the sump is at a low level based at least in part on the pressure reading, energize the pump assembly to circulate the wash fluid, obtain a sound signal generated during energization of the pump assembly using the microphone, determine that the wash fluid in the sump is not at a low level based at least in part on the sound signal, and provide a user notification that the pressure sensor is defective.

Inventors: Hu; Haitian (Louisville, KY), Hofmann; Adam (Louisville, KY)

Applicant: Haier US Appliance Solutions, Inc. (Wilmington, DE)

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Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to dishwasher appliances, and more particularly, to systems and methods for monitoring water levels in a dishwasher appliance.

BACKGROUND OF THE INVENTION

[0002] Dishwasher appliances generally include a tub that defines a wash chamber. Rack assemblies can be mounted within the wash chamber of the tub for receipt of articles for washing. Wash fluid (e.g., various combinations of water and detergent along with optional additives) may be introduced into the tub where it collects in a sump space at the bottom of the wash chamber. During wash and rinse cycles, a pump may be used to circulate wash fluid to spray assemblies within the wash chamber that can apply or direct wash fluid towards articles disposed within the rack assemblies in order to clean such articles. During a drain cycle, a pump may periodically discharge soiled wash fluid that collects in the sump space and the process may be repeated.

[0003] Certain conventional dishwasher appliances include pressure sensors to help determine the amount of water in the dishwasher. Notably, when water is filled in a dishwasher and a lower-than-normal reading is reported by the pressure sensor, the cause for the low reading may be due to a faulty pressure sensor or there may be an issue with the water filling process, such as low inlet water pressure. For example, a unit with a clogged pressure sensor generates a lower than expected water level after the completion of the fill, but the deviation from the intended range could have been caused by a malfunctioning water valve, low inlet water pressure, or any other problem unrelated to the clogged pressure sensor. Lack of knowledge as to the source of the low reading may result in incorrect diagnostics, improper maintenance, and consumer dissatisfaction.

[0004] Accordingly, a dishwasher appliance with an improved means for detecting the cause of low pressure readings is desired. More specifically, a dishwasher appliance that can identify a defective pressure sensor would be particularly beneficial.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

[0006] In one exemplary embodiment, a dishwasher appliance defining a vertical direction, a lateral direction, and a transverse direction is provided. The dishwasher appliance includes a wash tub that defines a wash chamber and a sump for collecting wash fluid, a pump assembly for selectively urging a flow of the wash fluid from the sump, a pressure sensor for monitoring a wash fluid level within the wash tub, a microphone for monitoring sound generated during operation of the dishwasher appliance, and a controller in operative communication with the pump assembly and the microphone. The controller is configured to obtain a pressure reading using the pressure sensor, determine that the wash fluid in the sump is at a low level based at least in part on the pressure reading, energize the pump assembly to circulate the wash fluid, obtain a sound signal generated during energization of the pump assembly using the microphone, determine that the wash fluid in the sump is not at a low level based at least in part on the sound signal, and provide a user notification that the pressure sensor is defective.

[0007] In another exemplary embodiment, a method of operating a dishwasher appliance is provided. The dishwasher appliance includes a pump assembly for selectively urging a flow of wash fluid from a sump, a pressure sensor for monitoring a wash fluid level within a wash tub, and

a microphone for monitoring sound generated during operation of the dishwasher appliance. The method includes obtaining a pressure reading using the pressure sensor, determining that the wash fluid in the sump is at a low level based at least in part on the pressure reading, energizing the pump assembly to circulate the wash fluid, obtaining a sound signal generated during energization of the pump assembly using the microphone, determining that the wash fluid in the sump is not at a low level based at least in part on the sound signal, and providing a user notification that the pressure sensor is defective.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. 1 provides a perspective view of a dishwasher appliance, including a dishwasher door according to an example embodiment of the present disclosure.

[0011] FIG. 2 provides a cross-sectional side view of the exemplary dishwashing appliance of FIG. 1 according to an example embodiment of the present disclosure.

[0012] FIG. 3 illustrates a method for operating a dishwasher appliance in accordance with one embodiment of the present disclosure.

[0013] FIG. 4 provides a method of detecting a faulty pressure sensor in a dishwasher appliance according to an example embodiment of the present subject matter.

[0014] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION

[0015] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0016] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. In addition, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). Furthermore, as used herein, terms of approximation, such as “approximately,” “substantially,” or “about,” refer to being within a ten percent margin of error.

[0017] FIGS. 1 and 2 depict an exemplary domestic dishwashing appliance or dishwasher **100** that may be configured in accordance with aspects of the present disclosure. For the particular embodiment of FIGS. 1 and 2, the dishwasher **100** includes a cabinet **102** having a tub **104** therein that defines a wash chamber **106**. As shown, tub **104** extends between a top **107** and a bottom **108** along a vertical direction V, between a pair of side walls **110** along a lateral direction L, and

between a front side **111** and a rear side **112** along a transverse direction T. Each of the vertical direction V, lateral direction L, and transverse direction T are mutually orthogonal to one another. [0018] The tub **104** includes a front opening **114** and a door **116** hinged at its bottom for movement between a normally closed vertical position (shown in FIG. 2), wherein the wash chamber **106** is sealed shut for washing operation, and a horizontal open position for loading and unloading of articles from the dishwasher **100**. According to exemplary embodiments, dishwasher **100** further includes a door closure mechanism or assembly **118** that is used to lock and unlock door **116** for accessing and sealing wash chamber **106**.

[0019] As illustrated in FIG. 2, tub side walls **110** may accommodate a plurality of rack assemblies. More specifically, guide rails **120** may be mounted to side walls **110** for supporting a lower rack assembly **122**, a middle rack assembly **124**, and an upper rack assembly **126**. As illustrated, upper rack assembly **126** is positioned at a top portion of wash chamber **106** above middle rack assembly **124**, which is positioned above lower rack assembly **122** along the vertical direction V. Each rack assembly **122**, **124**, **126** is adapted for movement between an extended loading position (not shown) in which the rack is substantially positioned outside the wash chamber **106**, and a retracted position (shown in FIGS. 1 and 2) in which the rack is located inside the wash chamber **106**. This is facilitated, for example, by rollers **128** mounted onto rack assemblies **122**, **124**, **126**, respectively. Although a guide rails **120** and rollers **128** are illustrated herein as facilitating movement of the respective rack assemblies **122**, **124**, **126**, it should be appreciated that any suitable sliding mechanism or member may be used according to alternative embodiments.

[0020] Some or all of the rack assemblies **122**, **124**, **126** are fabricated into lattice structures including a plurality of wires or elongated members **130** (for clarity of illustration, not all elongated members making up rack assemblies **122**, **124**, **126** are shown in FIG. 2). In this regard, rack assemblies **122**, **124**, **126** are generally configured for supporting articles within wash chamber **106** while allowing a flow of wash fluid to reach and impinge on those articles (e.g., during a cleaning or rinsing cycle). According to another exemplary embodiment, a silverware basket (not shown) may be removably attached to a rack assembly (e.g., lower rack assembly **122**) for placement of silverware, utensils, and the like, that are otherwise too small to be accommodated by rack **122**.

[0021] Dishwasher **100** further includes a plurality of spray assemblies for urging a flow of water or wash fluid onto the articles placed within wash chamber **106**. More specifically, as illustrated in FIG. 2, dishwasher **100** includes a lower spray arm assembly **134** disposed in a lower region **136** of wash chamber **106** and above a sump **138** so as to rotate in relatively close proximity to lower rack assembly **122**. Similarly, a mid-level spray arm assembly **140** is located in an upper region of wash chamber **106** and may be located below and in close proximity to middle rack assembly **124**. In this regard, mid-level spray arm assembly **140** may generally be configured for urging a flow of wash fluid up through middle rack assembly **124** and upper rack assembly **126**. Additionally, an upper spray assembly **142** may be located above upper rack assembly **126** along the vertical direction V. In this manner, upper spray assembly **142** may be configured for urging or cascading a flow of wash fluid downward over rack assemblies **122**, **124**, and **126**. As further illustrated in FIG. 2, upper rack assembly **126** may further define an integral spray manifold **144**, which is generally configured for urging a flow of wash fluid substantially upward along the vertical direction V through upper rack assembly **126**.

[0022] The various spray assemblies and manifolds described herein may be part of a fluid distribution system or fluid circulation assembly **150** for circulating water and wash fluid in the tub **104**. More specifically, fluid circulation assembly **150** includes a pump assembly **152** for circulating water or wash fluid (e.g., detergent, water, or rinse aid) in the tub **104**. For example, pump assembly **152** may include a pump motor that drives pump assembly **152** to circulate or discharge wash fluid. Pump assembly **152** may be located within sump **138** or within a machinery compartment located below sump **138** of tub **104**, as generally recognized in the art. Fluid circulation assembly **150** may include one or more fluid conduits or circulation piping for directing

water or wash fluid from pump assembly **152** to the various spray assemblies and manifolds. For example, as illustrated in FIG. 2, a primary supply conduit **154** may extend from pump assembly **152**, along rear **112** of tub **104** along the vertical direction V to supply wash fluid throughout wash chamber **106**.

[0023] As illustrated, primary supply conduit **154** is used to supply wash fluid to one or more spray assemblies (e.g., to mid-level spray arm assembly **140** and upper spray assembly **142**). However, it should be appreciated that according to alternative embodiments, any other suitable plumbing configuration may be used to supply wash fluid throughout the various spray manifolds and assemblies described herein. For example, according to another exemplary embodiment, primary supply conduit **154** could be used to provide wash fluid to mid-level spray arm assembly **140** and a dedicated secondary supply conduit (not shown) could be utilized to provide wash fluid to upper spray assembly **142**. Other plumbing configurations may be used for providing wash fluid to the various spray devices and manifolds at any location within dishwasher appliance **100**.

[0024] Each spray arm assembly **134**, **140**, **142**, integral spray manifold **144**, or other spray device may include an arrangement of discharge ports or orifices for directing wash fluid received from pump assembly **152** onto dishes or other articles located in wash chamber **106**. The arrangement of the discharge ports, also referred to as jets, apertures, or orifices, may provide a rotational force by virtue of wash fluid flowing through the discharge ports. Alternatively, spray arm assemblies **134**, **140**, **142** may be motor-driven, or may operate using any other suitable drive mechanism. Spray manifolds and assemblies may also be stationary. The resultant movement of the spray arm assemblies **134**, **140**, **142** and the spray from fixed manifolds provides coverage of dishes and other dishwasher contents with a washing spray. Other configurations of spray assemblies may be used as well. For example, dishwasher **100** may have additional spray assemblies for cleaning silverware, for scouring casserole dishes, for spraying pots and pans, for cleaning bottles, etc. One skilled in the art will appreciate that the embodiments discussed herein are used for the purpose of explanation only and are not limitations of the present subject matter.

[0025] In operation, pump assembly **152** draws wash fluid in from sump **138** and pumps it to a diverter assembly **156** (e.g., which may be positioned within sump **138** of dishwasher appliance **100**). Diverter assembly **156** may include a diverter disk (not shown) disposed within a diverter chamber **158** for selectively distributing the wash fluid to the spray arm assemblies **134**, **140**, **142** or other spray manifolds or devices. For example, the diverter disk may have a plurality of apertures that are configured to align with one or more outlet ports (not shown) at the top of diverter chamber **158**. In this manner, the diverter disk may be selectively rotated to provide wash fluid to the desired spray device.

[0026] According to an exemplary embodiment, diverter assembly **156** is configured for selectively distributing the flow of wash fluid from pump assembly **152** to various fluid supply conduits, only some of which are illustrated in FIG. 2 for clarity. More specifically, diverter assembly **156** may include four outlet ports (not shown) for supplying wash fluid to a first conduit for rotating lower spray arm assembly **134** in the clockwise direction, a second conduit for rotating lower spray arm assembly **134** in the counter-clockwise direction, a third conduit for spraying an auxiliary rack such as the silverware rack, and a fourth conduit for supply mid-level or upper spray assemblies **140**, **142** (e.g., such as primary supply conduit **154**).

[0027] The dishwasher **100** is further equipped with a controller **160** to regulate operation of the dishwasher **100**. The controller **160** may include one or more memory devices and one or more microprocessors, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with a cleaning cycle. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH. In one embodiment, the processor executes programming instructions stored in memory. The memory may be a separate component from the processor or may be included onboard within the processor. Alternatively, controller **160** may be constructed without using a microprocessor

(e.g., using a combination of discrete analog or digital logic circuitry, such as switches, amplifiers, integrators, comparators, flip-flops, AND gates, and the like) to perform control functionality instead of relying upon software.

[0028] The controller **160** may be positioned in a variety of locations throughout dishwasher **100**. In the illustrated embodiment, the controller **160** may be located within a control panel area **162** of door **116**, as shown in FIGS. **1** and **2**. In such an embodiment, input/output (“I/O”) signals may be routed between the control system and various operational components of dishwasher **100** along wiring harnesses that may be routed through the bottom of door **116**. Typically, the controller **160** may be in operative communication with a user interface panel **164** through which a user may select various operational features and modes and monitor progress of the dishwasher **100**. In one embodiment, the user interface **164** may represent a general purpose I/O (“GPIO”) device or functional block. In certain embodiments, the user interface **164** includes input components **166**, such as one or more of a variety of electrical, mechanical or electro-mechanical input devices including capacitive touch screens/buttons, rotary dials, push buttons, and touch pads. The user interface **164** may further include one or more display components **168**, such as a digital display device or one or more indicator light assemblies designed to provide operational feedback to a user. The user interface **164** may be in communication with the controller **160** via one or more signal lines or shared communication busses.

[0029] It should be appreciated that the invention is not limited to any particular style, model, or configuration of dishwasher **100**. The exemplary embodiment depicted in FIGS. **1** and **2** is for illustrative purposes only. For example, different locations may be provided for user interface **164**, different configurations may be provided for rack assemblies **122**, **124**, **126**, different spray arm assemblies **134**, **140**, **142** and spray manifold configurations may be used, and other differences may be applied while remaining within the scope of the present subject matter. Moreover, aspects of the present subject matter may be applied to other appliances as well, such as refrigerators, ovens, microwaves, etc.

[0030] Referring now generally to FIGS. **1** and **2**, door **116** will be described according to exemplary embodiments of the present subject matter. Although door **116** is described herein as being used with dishwasher **100**, it should be appreciated that door **116** or variations thereof may be used on any other suitable residential or commercial appliance. As described herein, door **116** may share a coordinate system with dishwasher **100**, e.g., when door **116** is in the closed position (e.g., as shown in FIG. **2**). Specifically, door **116** may define a vertical direction V, a lateral direction L, and a transverse direction T. Therefore, these directions may be used herein to refer to features of door **116** and its various components and sub-assemblies.

[0031] As shown, in the normally closed position, door **116** extends from a top end or top edge to a bottom end or bottom edge along the vertical direction V; from a front end to a rear end along the transverse direction T; and between two lateral ends along the lateral direction L. According to exemplary embodiments, door **116** may be formed from one or more exterior panels that define an interior chamber of door **116**. According to exemplary embodiments, the exterior panels of door **116** may be panels that are stamped from stainless steel or may be formed from any other suitably rigid material, such as thermoformed plastic, other metals, etc. In general, the exterior panels of door **116** may be assembled in any suitable manner, e.g., may be secured together using any suitable mechanical fastener, welding, snap-fit mechanisms, etc. In addition, it should be appreciated that an insulating material (not shown), such as fiberglass or foam insulation, may be positioned within door **116** to provide thermal and/or sound insulation to dishwasher **100**.

[0032] Referring still to FIGS. **1** and **2**, user interface panel **164** is positioned proximate the top edge of door **116** along the vertical direction V. In this manner, user interface panel **164** may be partially hidden below a countertop when dishwasher appliance **100** is installed below the countertop and door **116** is closed. Accordingly, dishwasher appliance **100** may be referred to as a “top control dishwasher appliance.” However, it should be appreciated that aspects of the present

subject matter may be used with dishwasher appliances having other configurations or any other suitable appliance. For example, user interface panel **164** may be alternately positioned on front face or front end of door **116**.

[0033] User interface panel **164** is positioned on door **116** such that a user can engage or interact with user interface panel **164**, e.g., to select operating cycles and parameters, activate/deactivate operating cycles, or adjust other operating parameters of dishwasher appliance **100**. User interface panel **164** may include a printed circuit board (not shown) that is positioned within door **116**. According to exemplary embodiments, printed circuit board may include or be operatively coupled to controller **160** and/or user interface panel **164**. In addition, user interface panel **164** may include or be operably coupled to one or more user inputs or touch buttons (e.g., identified generally herein as user inputs **166**) for receiving user input, providing user notifications, or illuminating to indicate cycle or operating status.

[0034] Specifically, according to the illustrated embodiment, user inputs **166** include a plurality of capacitive sensors that are mounted to user interface panel **164** and are operable to detect user inputs. For example, these capacitive sensors may be configured for triggering when a user touches the top edge of user interface panel **164** in a region associated with a particular user input **166**. In particular, these capacitive sensors can detect when a finger or another conductive material with a dielectric different than air contacts or approaches user interface panel **164**, along with the precise location, pressure, etc. of the finger interaction.

[0035] When a user touches the top edge of user interface panel **164** adjacent one of user inputs **166**, the associated capacitive sensors may be triggered and may communicate a corresponding signal to controller **160**. In such a manner, operations of dishwasher appliance **100** can be initiated and controlled. According to exemplary embodiments, the capacitive sensors may be distributed laterally on user interface panel **164**. It will be understood that other any suitable number, type, and position of capacitive sensors may be used while remaining within the scope of the present subject matter. Indeed, any suitable number, type, and configuration of user inputs **166** may be used while remaining within the scope of the present subject matter. User interface panel **164** may define a plurality of surfaces that are

[0036] intended to be illuminated for various purposes. For example, user inputs **166** may be illuminated by light sources to inform the user of the location of the button or to provide some other status indication. Notably, this illumination is typically achieved by directing a light beam along the vertical direction V onto the top edge of user interface panel **164**. Door **116** may further include a plurality of light sources or lighting devices that are configured for illuminating one or more surfaces of user interface panel **164**. It should be appreciated that these light sources may include any suitable number, type, configuration, and orientation of light sources mounted at any suitable location to illuminate status indicators or buttons in any suitable colors, sizes, patterns, etc. In other words, the light sources may be provided as any suitable number, type, position, and configuration of electrical light source(s), using any suitable light technology and illuminating in any suitable color. For example, the light sources may include one or more light emitting diodes (LEDs), which may each illuminate in a single color (e.g., white LEDs), or which may each illuminate in multiple colors (e.g., multi-color or RGB LEDs) depending on the control signal from controller **160**.

[0037] However, it should be appreciated that according to alternative embodiments, the light sources may include any other suitable traditional light bulbs or sources, such as halogen bulbs, fluorescent bulbs, incandescent bulbs, glow bars, a fiber light source, etc. Moreover, the light sources may be operably coupled (e.g., electrically coupled) to controller **160** or another suitable control board to facilitate activation or illumination of the light sources (e.g., to indicate a user input, state of the dishwasher appliance, state of the wash cycle, or any other relevant information to a user).

[0038] According to exemplary embodiments, user interface panel **164** may be any suitable

transparent or semitransparent feature for diffusing, directing, or otherwise transmitting light from a light source. For example, user interface panel **164** may be formed from a suitable transparent or translucent material configured to direct light energy, such as a dielectric material, such as glass or plastic, polycarbonate, polypropylene, polyacrylic, or any other suitable material.

[0039] In addition, user interface panel **164** may be a dead fronted panel. As used herein, the term “dead front” and the like is generally intended to refer to portions of a control panel which may be used as indicators, buttons, interactive control surfaces, or other user-interaction features without exposing the user to the operating side of the equipment or live parts and connections, i.e., lights, electrical connections, etc. For example, user interface panel **164** may include a transparent or translucent body and an opaque masking material that is selectively printed on the top edge of the translucent body to define capacitive touch buttons or user inputs **166**.

[0040] The opaque material may be deposited on the translucent body to define any suitable number, size, and configuration of illuminated features. These illuminated features may be shapes or include other forms such as symbols, words, etc. that are visible on user interface panel **164**. More specifically, when light sources are energized, capacitive touch buttons or user inputs **166** on the top edge may be illuminated. Thus, the dead fronted the top edge may be the surface that is contacted for controlling dishwasher appliance **100** or which may be illuminated for purposes of indicating operating status or other conditions to the user of the dishwasher appliance **100**.

[0041] Referring still to FIG. **1**, a schematic diagram of an external communication system **180** will be described according to an exemplary embodiment of the present subject matter. In general, external communication system **180** is configured for permitting interaction, data transfer, and other communications between dishwasher appliance **100** and one or more external devices. For example, this communication may be used to provide and receive operating parameters, user instructions or notifications, performance characteristics, user preferences, or any other suitable information for improved performance of dishwasher appliance **100**. In addition, it should be appreciated that external communication system **180** may be used to transfer data or other information to improve performance of one or more external devices or appliances and/or improve user interaction with such devices.

[0042] For example, external communication system **180** permits controller **160** of dishwasher appliance **100** to communicate with a separate device external to dishwasher appliance **100**, referred to generally herein as an external device **182**. As described in more detail below, these communications may be facilitated using a wired or wireless connection, such as via a network **184**. In general, external device **182** may be any suitable device separate from dishwasher appliance **100** that is configured to provide and/or receive communications, information, data, or commands from a user. In this regard, external device **182** may be, for example, a personal phone, a smartphone, a tablet, a laptop or personal computer, a wearable device, a smart home system, or another mobile or remote device.

[0043] In addition, a remote server **186** may be in communication with dishwasher appliance **100** and/or external device **182** through network **184**. In this regard, for example, remote server **186** may be a cloud-based server **186**, and is thus located at a distant location, such as in a separate state, country, etc. According to an exemplary embodiment, external device **182** may communicate with a remote server **186** over network **184**, such as the Internet, to transmit/receive data or information, provide user inputs, receive user notifications or instructions, interact with or control dishwasher appliance **100**, etc. In addition, external device **182** and remote server **186** may communicate with dishwasher appliance **100** to communicate similar information.

[0044] In general, communication between dishwasher appliance **100**, external device **182**, remote server **186**, and/or other user devices or appliances may be carried using any type of wired or wireless connection and using any suitable type of communication network, non-limiting examples of which are provided below. For example, external device **182** may be in direct or indirect communication with dishwasher appliance **100** through any suitable wired or wireless

communication connections or interfaces, such as network **184**. For example, network **184** may include one or more of a local area network (LAN), a wide area network (WAN), a personal area network (PAN), the Internet, a cellular network, any other suitable short- or long-range wireless networks, etc. In addition, communications may be transmitted using any suitable communications devices or protocols, such as via Wi-Fi®, Bluetooth®, Zigbee®, wireless radio, laser, infrared, Ethernet type devices and interfaces, etc. In addition, such communication may use a variety of communication protocols (e.g., TCP/IP, HTTP, SMTP, FTP), encodings or formats (e.g., HTML, XML), and/or protection schemes (e.g., VPN, secure HTTP, SSL).

[0045] External communication system **180** is described herein according to an exemplary embodiment of the present subject matter. However, it should be appreciated that the exemplary functions and configurations of external communication system **180** provided herein are used only as examples to facilitate description of aspects of the present subject matter. System configurations may vary, other communication devices may be used to communicate directly or indirectly with one or more associated appliances, other communication protocols and steps may be implemented, etc. These variations and modifications are contemplated as within the scope of the present subject matter.

[0046] As noted briefly above, pressure sensors may be used in dishwashers to help determine the amount of water in the dishwasher. Accordingly, as illustrated, dishwasher appliance **100** includes a pressure sensor **190** operably coupled to sump **138** for measuring a pressure of wash fluid within sump **138**, e.g., to facilitate wash fluid level detection. In general, pressure sensor **190** may be any sensor suitable for determining a water level within sump **138** based on pressure readings. For example, pressure sensor **190** may be a piezoelectric pressure sensor and thus may include an elastically deformable plate and a piezoresistor mounted on the elastically deformable plate. However, it should be appreciated that according to alternative embodiments, pressure sensor **190** may be any type of pressure sensor that is fluidly coupled to sump **138** in any other suitable manner for obtaining sump pressures to facilitate water level detection.

[0047] Notably, when water is filled in a dishwasher and a lower-than-normal reading is reported by the pressure sensor, the cause for the low reading may be due to a faulty pressure sensor or there may be an issue with the water filling process, such as low inlet water pressure. For example, a unit with a clogged pressure sensor may generate a lower than expected water level after the completion of the fill, but the deviation from the intended range could have been caused by a malfunctioning water valve, low inlet water pressure, or any other problem unrelated to the clogged pressure sensor. Lack of knowledge as to the source of the low reading may result in incorrect diagnostics, improper maintenance, and consumer dissatisfaction.

[0048] Accordingly, as explained in more detail below, dishwasher appliance **100** may utilize a microphone to help diagnose a faulty pressure sensor or other operational issues with dishwasher appliance **100**. Specifically, dishwasher appliance **100** may include a microphone **192** that is positioned at any suitable location within the vicinity of dishwasher appliance **100** for monitoring sound generated by dishwasher appliance **100**. For example, according to the illustrated embodiment, microphone **192** may be positioned within or proximate to sump **138**. In general, microphone **192** may be used for monitoring the sound waves, noises, or other vibrations generated during the operation of dishwasher appliance **100**. For example, microphone **192** may be one or more microphones, acoustic detection devices, vibration sensors, or any other suitable acoustic transducers that are positioned at one or more locations in or around dishwasher appliance **100**.

[0049] As best shown in FIG. 2, dishwasher appliance **100** may further include a water supply **194** that is generally configured for dispensing a flow of water into wash tub **104** to facilitate appliance operation. For example, water supply **194** generally includes a spout that is fluidly coupled to sump **138** and is configured for directing a flow of fluid into wash tub **104**. In addition, a water supply valve **196** may regulate the flow of water from water supply **194** (such as a municipal water supply) into wash tub **104**. It should be appreciated that water supply **194** and water supply valve

194 may be positioned at any other suitable location within cabinet **102**. Other systems and methods for supplying water are possible and within the scope of the present subject matter.

[0050] Now that the construction of dishwasher appliance **100** according to exemplary embodiments have been presented, an exemplary method **200** of operating a dishwasher appliance will be described. Although the discussion below refers to the exemplary method **200** of operating dishwasher appliance **100**, one skilled in the art will appreciate that the exemplary method **200** is applicable to the operation of a variety of other dishwasher appliances and motors on any other suitable appliance.

[0051] In exemplary embodiments, the various method steps as disclosed herein may be performed by controller **160** or a separate, dedicated controller. In this regard, as described herein, controller **160** of dishwasher appliance **100** may implement all steps of method **200**. However, it should be appreciated that according to alternative embodiments, controller **160** may offload the analysis described herein, e.g., by sending pressure readings and sound signals to remote server **186** for analysis. Other distributed analysis arrangements are possible and within the scope of the present subject matter.

[0052] Referring now to FIG. **3**, method **200** includes, at step **210**, obtaining a pressure reading using a pressure sensor mounted in a sump of a dishwasher appliance. In this regard, continuing the example from above, pressure sensor **190** may be used to periodically or continuously monitor the water pressure and water level within wash tub **104**. According to example embodiments, method **200** may further include operating water supply **194** and water supply valve **196** to perform a fill cycle to add water into wash tub **104**. In this regard, water supply valve **196** may be open for a predetermined amount of time to dispense a target volume of water for a specific wash cycle of dishwasher appliance **100**. According to example embodiments, the pressure reading obtained at step **210** may be obtained after completion of the fill cycle, e.g., when the target volume of water is presumably within wash tub **104**.

[0053] Step **220** may include determining that the wash fluid in the sump is at a low level based at least in part on the pressure reading. In this regard, controller **160** may be programmed with a predetermined pressure range or threshold that is associated with a properly filled wash tub. If the pressure sensor indicates that the sump pressure is below that target threshold or outside the target range, controller **160** may determine that the wash fluid in sump **138** is low. As noted above, if pressure sensor **190** is faulty, there may be sufficient water in wash tub **104** and it may be undesirable to implement a responsive action based on the erroneous pressure reading. By contrast, if pressure sensor **190** is not faulty, the water level may be low due to a variety of other factors, such as low water pressure, a faulty water supply valve **196**, etc.

[0054] Accordingly, method **200** may further include steps for verifying the reading of pressure sensor **190** and for diagnosing other operational issues with dishwasher appliance **100**. For example, step **230** may include energizing a pump assembly to circulate the water or wash fluid within wash tub **104**. In this regard, continuing the example from above, after completion of the fill cycle, controller **160** may operate pump assembly **152** with the intention of circulation wash water, discharging wash water, etc. Notably, the sound generated by operation of pump assembly **152** may vary depending on whether there is a sufficient or target volume of wash fluid within sump **138**. For example, operation of pump assembly **152** may be relatively quiet if the target volume of wash fluid is being circulated, whereas the volume may increase substantially when there is a low level of wash fluid within sump **132**, e.g., due to air being sucked into pump assembly **152**.

[0055] Therefore, it may be desirable to monitor the sound generated by pump assembly **152**. Thus, step **240** may generally include obtaining a sound signal generated during energization of the pump assembly using a microphone mounted in the sump or at another suitable location on the dishwasher appliance. In this regard, during and immediately after energizing pump assembly **152**, controller **160** may use microphone **192** to monitor sounds or noises generated within and around dishwasher appliance **100**. By monitoring the sound generated, controller **160** may identify issues

with pump assembly, water supply **194**, water supply valve **196**, and/or pressure sensor **190**.

[0056] In general, the analysis of the sound signal may be made based on the amplitude of sound (e.g., in decibels) falling within or outside of predetermined ranges, thresholds, etc. In addition, or alternatively, method **200** may include identifying unique sound signatures or audio profiles within the sound signal. These sound signatures may be associated with specific events, such as the sucking of air into pump assembly **152**. It should be appreciated that the analysis of the sound signal may be performed locally by controller **160** or remotely, e.g., on remote server **186**. Other methods of sound analysis are possible and within the scope of the present subject matter.

[0057] Step **250** may generally include determining that the wash fluid in the sump is not at a low level based at least in part on the sound signal. In this regard, if the sound signal generated by operating pump assembly **152** falls within the normal operating range or does not exceed a predetermined sound threshold, this may indicate that the amount of wash fluid within sump **138** is sufficient, e.g., the water level exceeds the low level detected by pressure sensor **190**. Accordingly, this may be indicative of a situation where the pressure sensor is defective or faulty, e.g., as indicated by the conflicting determinations made using the two sensors.

[0058] Accordingly, step **260** may include providing a user notification that the pressure sensor is defective. This user notification may be communicated to the user in any suitable manner. For example, the user notification may be provided through a user interface panel (such as user interface panel **164**), e.g., by illuminating a faulty sensor light indicator. According to still other embodiments, the user notification regarding the pressure sensor failure may be communicated directly to the user through external device **182** (e.g., such as the user's cell phone) via network **184**. According to still other embodiments, the user notification may be provided to a smart speaker, to another connected appliance, or any other suitable device. In addition, according to example embodiments, method **200** may include communicating with a service technician, scheduling a maintenance service, ordering a new pressure sensor or part, etc.

[0059] According to still other embodiments, method **200** may include steps for determining that the water supply or water supply valves are defective. In this regard, method **200** may include determining that the wash fluid in the sump is at a low level based on the pressure reading and confirming that the wash fluid in the sump is at the low level based on the sound signal. Because analysis of these two signals both indicate that the water level is low, this may be indicative of a faulty water supply valve **196**, a low water pressure of the water supply **194**, a tub leak, or other operational issues with dishwasher appliance **100**. Accordingly, method **200** may include providing a user notification or taking other corrective action to address these issues.

[0060] Referring now briefly to FIG. **4**, a method **300** for detecting a faulty pressure sensor or a water filling issue will be described according to an example embodiment of the present subject matter. It should be appreciated that some or all of the steps of method **300** may be the same or similar to steps of method **200**, and like reference numerals may be used to refer to the same or similar steps herein. Specifically, step **302** may include determining whether the fill cycle has been completed. This would indicate that a target volume of water is in sump **138** if dishwasher appliance **100** and water supply **194** are operating correctly. If the fill cycle is complete, step **304** may include determining that pressure sensor **190** senses that the water level is below a predetermined low water level threshold.

[0061] If the pressure reading is acceptable, step **306** ends method **300**, otherwise, step **308** includes monitoring the sound of a circulation pump and sending the sound signal to the cloud for analysis. Step **310** includes the performance of in-cloud analytics and step **312** includes determining whether the sound signal matches normal operation of the pump assembly. If the sound signal is normal or expected, this may indicate a faulty pressure sensor at step **314**. Alternatively, if the sound signal indicates that the water level is low (e.g., thus matching the pressure reading), this may indicate a water filling issue at step **316**.

[0062] FIGS. **3** and **4** depict exemplary control methods having steps performed in a particular

order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that the steps of any of the methods discussed herein can be adapted, rearranged, expanded, omitted, or modified in various ways without deviating from the scope of the dishwasher appliance **100** as an example, it should be appreciated that these methods may be applied to other dishwasher appliance, or other appliances such as washing machine appliances.

[0063] As explained above, aspects of the present subject matter are generally directed to a cloud-based dishwasher with a self-diagnostic feature for detecting a faulty pressure sensor. The self-diagnostic feature may utilize an innovative algorithm that integrates circulation pump's sound analytics with pressure sensor's readings after a fill cycle to determine if there is a defective pressure sensor or a water filling issue. The algorithm may determine the sound difference between normal operation and low water operation when the dishwasher runs without sufficient water; where the circulation pump pushes the available water to spray arms resulting in a loud noise caused by the water/air mixture within the pump due to the air drawn before the water has a chance to return to the pump inlet. The occurrence of this loud noise may correspond to a specific threshold value unique to each dishwasher model, which serves as a target parameter for the algorithm. The algorithm may determine the root cause of low-pressure sensor reading after a fill cycle designed to detect a lower-than-normal reading on the dishwasher's pressure sensor after the fill cycle. To verify the proper priming of the circulation pump, the algorithm may utilize a sound reading generated when the pump is activated by employing a microphone to distinguish between the sounds associated with normal operation and those indicative of low water operation. When the pressure sensor registers a low reading while the sound reading indicates an abnormal operation, this may indicate a problem with the water filling process. Consequently, an alert may be promptly sent to the user, notifying them of the water filling issue. Conversely, if the pressure sensor indicates low pressure and the sound reading indicates a normal operation, this may suggest a malfunctioning pressure sensor. In such cases, an alert notifying the user of the faulty pressure sensor may be generated.

[0064] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A dishwasher appliance defining a vertical direction, a lateral direction, and a transverse direction, the dishwasher appliance comprising: a wash tub that defines a wash chamber and a sump for collecting wash fluid; a pump assembly for selectively urging a flow of the wash fluid from the sump; a pressure sensor for monitoring a wash fluid level within the wash tub; a microphone for monitoring sound generated during operation of the dishwasher appliance; and a controller in operative communication with the pump assembly and the microphone, the controller being configured to: obtain a pressure reading using the pressure sensor; determine that the wash fluid in the sump is at a low level based at least in part on the pressure reading; energize the pump assembly to circulate the wash fluid; obtain a sound signal generated during energization of the pump assembly using the microphone; determine that the wash fluid in the sump is not at a low level based at least in part on the sound signal; and provide a user notification that the pressure sensor is defective.

2. The dishwasher appliance of claim 1, further comprising a water supply for selectively adding the wash fluid into the wash tub, wherein the controller is further configured to operate the water supply to perform a fill cycle to add the wash fluid into the wash tub, wherein the pressure reading and the sound signal are obtained after completion of the fill cycle.
3. The dishwasher appliance of claim 2, wherein the controller is further configured to: determine that the wash fluid in the sump is at a low level based at least in part on the pressure reading; confirm that the wash fluid in the sump is at the low level based at least in part on the sound signal; and provide a user notification that the water supply is defective.
4. The dishwasher appliance of claim 1, wherein confirming that the wash fluid in the sump is at the low level based at least in part on the sound signal comprises: analyzing the sound signal to detect a low level sound signature associated with a low fill level.
5. The dishwasher appliance of claim 4, wherein the low level sound signature is associated with air being drawn into the pump assembly.
6. The dishwasher appliance of claim 1, wherein confirming that the wash fluid in the sump is at the low level based at least in part on the sound signal comprises: determining that an amplitude of the sound signal exceeds predetermined decibel threshold.
7. The dishwasher appliance of claim 1, wherein determining that the wash fluid in the sump is at the low level based at least in part on the pressure reading comprises: determining that the pressure reading is below a predetermined threshold value after completion of a fill cycle.
8. The dishwasher appliance of claim 1, wherein the microphone is a piezoelectric microphone.
9. The dishwasher appliance of claim 1, wherein the pressure sensor and the microphone are positioned within the sump of the dishwasher appliance.
10. The dishwasher appliance of claim 1, wherein the controller is configured to: transmit the sound signal to a remote server for analysis.
11. The dishwasher appliance of claim 1, wherein the user notification is provided through a user interface panel.
12. The dishwasher appliance of claim 1, wherein the controller is in operative communication with a remote device through an external network, and wherein the user notification is provided through the remote device.
13. The dishwasher appliance of claim 12, wherein the remote device comprises at least one of a mobile phone or a smart speaker.
14. A method of operating a dishwasher appliance, the dishwasher appliance comprising a pump assembly for selectively urging a flow of wash fluid from a sump, a pressure sensor for monitoring a wash fluid level within a wash tub, and a microphone for monitoring sound generated during operation of the dishwasher appliance, the method comprising: obtaining a pressure reading using the pressure sensor; determining that the wash fluid in the sump is at a low level based at least in part on the pressure reading; energizing the pump assembly to circulate the wash fluid; obtaining a sound signal generated during energization of the pump assembly using the microphone; determining that the wash fluid in the sump is not at a low level based at least in part on the sound signal; and providing a user notification that the pressure sensor is defective.
15. The method of claim 14, wherein the dishwasher appliance further comprises a water supply for selectively adding the wash fluid into the wash tub and the method further comprises operating the water supply to perform a fill cycle to add the wash fluid into the wash tub, wherein the pressure reading and the sound signal are obtained after completion of the fill cycle.
16. The method of claim 15, further comprising: determining that the wash fluid in the sump is at a low level based at least in part on the pressure reading; confirming that the wash fluid in the sump is at the low level based at least in part on the sound signal; and providing a user notification that the water supply is defective.
17. The method of claim 14, wherein confirming that the wash fluid in the sump is at the low level based at least in part on the sound signal comprises: analyzing the sound signal to detect a low

level sound signature associated with a low fill level.

18. The method of claim 17, wherein the low level sound signature is associated with air being drawn into the pump assembly.

19. The method of claim 14, wherein confirming that the wash fluid in the sump is at the low level based at least in part on the sound signal comprises: determining that an amplitude of the sound signal exceeds predetermined decibel threshold.

20. The method of claim 14, wherein determining that the wash fluid in the sump is at the low level based at least in part on the pressure reading comprises: determining that the pressure reading is below a predetermined threshold value after completion of a fill cycle.
